13 次单位根的根式解

参数与标准素分解

$p = 13$，$h = (p-1)/2 = 6$，模 13 的最小原根 $g = 2$。

h 的标准素分解：$h = 3 \times 2$，子群链共 2 层。

$m_0 = 1$，$m_1 = 3$，$m_2 = 6$。

各层子群 $H_j = \{\pm g^{m_j t}\}$，其中 H_0 是全部 12 个非零剩余：

$$H_1 = \{1, 5, 8, 12\}$$

$$H_2 = \{1, 12\}$$

各层周期 $\eta_a^{(j)} = \sum_{b \in H_j} \zeta^{ab}$（$a$ 取陪集代表）。

各层周期与分解表

第 0 层：1 个周期

H_0 是全部 12 个非零剩余，故 $\eta_1^{(0)} = -1$。

第 1 层：3 个周期

$$\eta_1^{(1)} = \zeta^1 + \zeta^5 + \zeta^8 + \zeta^{12}$$

$$\eta_2^{(1)} = \zeta^2 + \zeta^3 + \zeta^{10} + \zeta^{11}$$

$$\eta_4^{(1)} = \zeta^4 + \zeta^6 + \zeta^7 + \zeta^9$$

第 2 层：6 个周期

$$\eta_1^{(2)} = \zeta^1 + \zeta^{12}$$

$$\eta_2^{(2)} = \zeta^2 + \zeta^{11}$$

$$\eta_3^{(2)} = \zeta^3 + \zeta^{10}$$

$$\eta_4^{(2)} = \zeta^4 + \zeta^9$$

$$\eta_5^{(2)} = \zeta^5 + \zeta^8$$

$$\eta_6^{(2)} = \zeta^6 + \zeta^7$$

分解表：每个母周期拆成 ℓ_j 个子周期

$$\eta_1^{(0)} = \eta_1^{(1)} + \eta_2^{(1)} + \eta_4^{(1)}$$

（第 1 层，$\ell = 3$。）

$$\eta_1^{(1)} = \eta_1^{(2)} + \eta_5^{(2)}$$

$$\eta_2^{(1)} = \eta_2^{(2)} + \eta_3^{(2)}$$

$$\eta_4^{(1)} = \eta_4^{(2)} + \eta_6^{(2)}$$

（第 2 层，$\ell = 2$。）

辅助单位根

$\ell = 2$ 时 $\omega_2^1 = -1$。
$\ell \geq 3$ 时按主支：

$$\omega_\ell^k = \frac{\Lambda_{\bar{k}}^{(\ell)}}{2} + \frac{\varepsilon_k}{2}\sqrt{2 - \Lambda_{2\bar{k}}^{(\ell)}}\, i$$

其中 $\varepsilon_k = \operatorname{sgn}\sin(2\pi k/\ell)$，$\bar{k} = \min(k, \ell - k)$。

辅助问题 $p = 3$（供 ω_3^k 使用）

$h = 1$，$g = 2$，层阶 1。

$$\Lambda_1^{(3)} = -1$$

$\omega_3^k = \frac{1}{2}\Lambda_{\bar{k}}^{(3)} + \frac{\varepsilon_k}{2}\sqrt{2 - \Lambda_{2\bar{k}}^{(3)}}\, i$。

逐层开方

每一步把母周期 $\eta_a^{(j-1)}$ 分解成 ℓ 个子周期 $\beta_t^{(a)}$。
预解式取离散 Fourier 变换：

$$\theta_{a,k}^{(j)} = \sum_{t=0}^{\ell-1} \omega_\ell^{-kt}\, \beta_t^{(a)}$$

$$\left(\theta_{a,k}^{(j)}\right)^\ell = u_{a,k}^{(j)} = \tau^k\left(u_{a,1}^{(j)}\right)$$

$$\eta_{b(t)}^{(j)} = \frac{1}{\ell}\left(\eta_a^{(j-1)} + \sum_{k=1}^{\ell-1} \omega_\ell^{kt}\, \theta_{a,k}^{(j)}\right)$$

（τ 把 ω_ℓ 送到 ω_ℓ^k；$b(t)$ 为第 t 个子周期。）
开方一律取正实主支；预解式的分支由因子 ω_ℓ^s 记录（见报告里的分支表）。

第 1 步（$\ell = 3$，1 个母周期）

$$u_{1,1}^{(1)} = \frac{-65 + 39i\sqrt{3}}{2}$$

$$\theta_{1,1}^{(1)} = \frac{\left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}}}{4}$$

第 2 步（$\ell = 2$，3 个母周期）

$$u_{1,1}^{(2)} = \frac{26 + \left(2 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} + \left(2 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}}}{6}$$

$$\theta_{1,1}^{(2)} = \frac{\sqrt{156 + \left(12 + 6i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} + \left(12 - 6i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}}}}{6}$$

$$u_{2,1}^{(2)} = \frac{52 + \left(1 + 3i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} + \left(1 - 3i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}}}{12}$$

$$\theta_{2,1}^{(2)} = \frac{\sqrt{156 + \left(3 + 9i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} + \left(3 - 9i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}}}}{6}$$

$$u_{4,1}^{(2)} = \frac{52 + \left(-5 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} + \left(-5 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}}}{12}$$

$$\theta_{4,1}^{(2)} = \frac{\sqrt{156 + \left(-15 + 3i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} + \left(-15 - 3i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}}}}{6}$$

实部数值

$$\Lambda_1 = 2\cos\frac{2\pi\cdot 1}{13} = \frac{-4 + \left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} + \left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} + 2\sqrt{156 + \left(12 + 6i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} + \left(12 - 6i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}}}}{24}$$

$$\Lambda_2 = 2\cos\frac{2\pi\cdot 2}{13} = \frac{-2 + \sqrt[3]{-260 + 156i\sqrt{3}} + \sqrt[3]{-260 - 156i\sqrt{3}} + \sqrt{156 + \left(3 + 9i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} + \left(3 - 9i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}}}}{12}$$

$$\Lambda_3 = 2\cos\frac{2\pi\cdot 3}{13} = \frac{-2 + \sqrt[3]{-260 + 156i\sqrt{3}} + \sqrt[3]{-260 - 156i\sqrt{3}} - \sqrt{156 + \left(3 + 9i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} + \left(3 - 9i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}}}}{12}$$

$$\Lambda_4 = 2\cos\frac{2\pi\cdot 4}{13} = \frac{-4 + \left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} + \left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} + 2\sqrt{156 + \left(-15 + 3i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} + \left(-15 - 3i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}}}}{24}$$

$$\Lambda_5 = 2\cos\frac{2\pi\cdot 5}{13} = \frac{-4 + \left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} + \left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} - 2\sqrt{156 + \left(12 + 6i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} + \left(12 - 6i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}}}}{24}$$

$$\Lambda_6 = 2\cos\frac{2\pi\cdot 6}{13} = \frac{-4 + \left(-1 + i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}} + \left(-1 - i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} - 2\sqrt{156 + \left(-15 + 3i\sqrt{3}\right)\sqrt[3]{-260 - 156i\sqrt{3}} + \left(-15 - 3i\sqrt{3}\right)\sqrt[3]{-260 + 156i\sqrt{3}}}}{24}$$

全部单位根

最终形式：

$$\omega_n^t = \frac{\Lambda_{\bar{t}}}{2} + \frac{\varepsilon_t}{2}\sqrt{2-\Lambda_{\overline{2t}}}\,i, \qquad \Lambda_{\bar{t}} = 2\cos\frac{2\pi\bar{t}}{n}$$

其中 $\bar{t} = \min(t, n-t)$，$\varepsilon_t = \operatorname{sgn}\sin(2\pi t/n)$。

以下逐一写开（全部为具体根式）：

$$\omega_{13}^{1} = \frac{-4+(-1+i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(-1-i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+2\sqrt{156+(12+6i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(12-6i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}}+4i\sqrt{78-3\sqrt[3]{-260+156i\sqrt{3}}-3\sqrt[3]{-260-156i\sqrt{3}}-3\sqrt{156+(3+9i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+(3-9i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}}}}{48}$$

$$\omega_{13}^{2} = \frac{-2+\sqrt[3]{-260+156i\sqrt{3}}+\sqrt[3]{-260-156i\sqrt{3}}+\sqrt{156+(3+9i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+(3-9i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}}+i\sqrt{312+(6+6i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(6-6i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}-12\sqrt{156+(-15+3i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(-15-3i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}}}}{24}$$

$$\omega_{13}^{3} = \frac{-2+\sqrt[3]{-260+156i\sqrt{3}}+\sqrt[3]{-260-156i\sqrt{3}}-\sqrt{156+(3+9i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+(3-9i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}}+i\sqrt{312+(6+6i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(6-6i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+12\sqrt{156+(-15+3i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(-15-3i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}}}}{24}$$

$$\omega_{13}^{4} = \frac{-4+(-1+i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+(-1-i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+2\sqrt{156+(-15+3i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(-15-3i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}}+2i\sqrt{312+(6+6i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+(6-6i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+12\sqrt{156+(12+6i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(12-6i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}}}}{48}$$

$$\omega_{13}^{5} = \frac{-4+(-1+i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(-1-i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}-2\sqrt{156+(12+6i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(12-6i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}}+4i\sqrt{78-3\sqrt[3]{-260+156i\sqrt{3}}-3\sqrt[3]{-260-156i\sqrt{3}}+3\sqrt{156+(3+9i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+(3-9i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}}}}{48}$$

$$\omega_{13}^{6} = \frac{-4+(-1+i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+(-1-i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}-2\sqrt{156+(-15+3i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(-15-3i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}}+2i\sqrt{312+(6+6i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+(6-6i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}-12\sqrt{156+(12+6i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(12-6i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}}}}{48}$$

$$\omega_{13}^{7} = \frac{-4+(-1+i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+(-1-i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}-2\sqrt{156+(-15+3i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(-15-3i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}}-2i\sqrt{312+(6+6i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+(6-6i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}-12\sqrt{156+(12+6i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(12-6i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}}}}{48}$$

$$\omega_{13}^{8} = \frac{-4+(-1+i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(-1-i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}-2\sqrt{156+(12+6i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(12-6i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}}-4i\sqrt{78-3\sqrt[3]{-260+156i\sqrt{3}}-3\sqrt[3]{-260-156i\sqrt{3}}+3\sqrt{156+(3+9i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+(3-9i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}}}}{48}$$

$$\omega_{13}^{9} = \frac{-4+(-1+i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+(-1-i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+2\sqrt{156+(-15+3i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(-15-3i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}}-2i\sqrt{312+(6+6i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+(6-6i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+12\sqrt{156+(12+6i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(12-6i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}}}}{48}$$

$$\omega_{13}^{10} = \frac{-2+\sqrt[3]{-260+156i\sqrt{3}}+\sqrt[3]{-260-156i\sqrt{3}}-\sqrt{156+(3+9i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+(3-9i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}}-i\sqrt{312+(6+6i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(6-6i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+12\sqrt{156+(-15+3i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(-15-3i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}}}}{24}$$

$$\omega_{13}^{11} = \frac{-2+\sqrt[3]{-260+156i\sqrt{3}}+\sqrt[3]{-260-156i\sqrt{3}}+\sqrt{156+(3+9i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+(3-9i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}}-i\sqrt{312+(6+6i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(6-6i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}-12\sqrt{156+(-15+3i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(-15-3i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}}}}{24}$$

$$\omega_{13}^{12} = \frac{-4+(-1+i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(-1-i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+2\sqrt{156+(12+6i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}+(12-6i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}}-4i\sqrt{78-3\sqrt[3]{-260+156i\sqrt{3}}-3\sqrt[3]{-260-156i\sqrt{3}}-3\sqrt{156+(3+9i\sqrt{3})\sqrt[3]{-260+156i\sqrt{3}}+(3-9i\sqrt{3})\sqrt[3]{-260-156i\sqrt{3}}}}}{48}$$